

Foundation (Jalapeno)

- Q1.** Factor the expression: $x^2 - 4$
(A) $(x + 2)(x - 2)$
(B) $x(x - 4)$
(C) $x(x + 4)$
(D) $2(x - 2)$
(E) $2(x + 2)$
- Q2.** If a stock's price increases by $2^2 - 1^2$ per day, how much does it increase by in dollars per day?
(A) 3
(B) 4
(C) 5
(D) 6
(E) 7
- Q3.** Factor the expression: $49 - 9$
(A) $5 \cdot 8$
(B) $4 \cdot 10$
(C) $7^2 - 3^2$
(D) $8^2 - 4^2$
(E) $6^2 - 2^2$
- Q4.** A company's savings account earns interest of $x^2 - 9$ dollars per year. If $x = 5$, how much does the account earn?
(A) 16
(B) 20
(C) 25
(D) 30
(E) 36
- Q5.** Factor the expression: $25 - 4y^2$
(A) $(5 - 2y)(5 + 2y)$
(B) $(5 + 2y)(5 - 2y)$
(C) $(25 + 4y^2)$
(D) $(25 - 4y)$
(E) $(25 + 4y)$
- Q6.** A tax consultant uses the formula $x^2 - 4$ to calculate the tax on a certain investment. If $x = 3$, what is the tax?
(A) 5
(B) 3
(C) 1
(D) 0
(E) -1
- Q7.** Factor the expression: $x^2 - 16$
(A) $(x + 4)(x - 4)$
(B) $(x - 4)(x + 4)$
(C) $(x^2 + 16)$
(D) $(x + 16)$
(E) $(x - 16)$
- Q8.** A stock's price is calculated by the formula $x^2 - 9$. If the price is 16, what is the value of x ?
(A) 5
(B) -5
(C) 4
(D) -4
(E) 3

Open Ended Questions (FRQ)

- Q9.** A company's profit from the sale of two products, A and B, is given by the expression $x^2 - 4y^2$. Factor the expression and explain its significance in the context of the company's finances. Provide your answer in 5 lines or less.
- Q10.** A savings account earns an interest of $25 - x^2$ dollars per year. Factor the expression and determine the value of x if the interest earned is 16 dollars. Provide your answer in 5 lines or less.

Chef's Tips

- Tip 1: Ensure students understand the difference of squares formula.
- Tip 2: Encourage students to apply the formula in various contexts.
- Tip 3: Remind students to check their answers by multiplying the factors.